

GNIOT Institute of Management Studies – PGDM (Batch 2020-22)**(TRIMESTER – IV) END TERM EXAMINATION, OCTOBER - 2021****FINANCIAL DERIVATIVES****[Time: 2 Hours 30 Minutes]****[Maximum Marks: 100]****INSTRUCTIONS:**

- (i) There are **THREE** sections in this question paper.
- (ii) Attempt questions from each section as per the directions.
- (iii) Make suitable assumptions wherever necessary.

SECTION-A**(Short Answer Type Questions, do not exceed 100 words)****Q.1. Attempt all parts of the following:****(10×3=30)**

- (a) Explain the term ‘financial derivative’. What are its important features?
- (b) Explain Swaps and their features.
- (c) Derivatives are the tools to hedge risk and speculate also.
- (d) Write a note on Currency Swap.
- (e) What do you mean by hedging? Discuss with suitable examples.
- (f) What is Exchange- traded derivatives?
- (g) Explain future contract.
- (h) Differentiate between put option and call option.
- (i) What is Hedge?
- (j) Explain the valuation of forward contract.

SECTION-B**(Long Answer Type Questions)****Attempt all questions of the following:****(4×10=40)**

Q.2. (a) Explain the term ‘Arbitrage’. Assume stock DEF trades New York Stock Exchange (NYSE) and the Tokyo Stock Exchange (TSE). The stock currently trades on the NYSE for \$ 35 and on the TSE for ¥ 2950. Given the current exchange ratio is 0.0105\$/ ¥, determine if an arbitrage profit is possible.

OR

(b) What is financial swap? Discuss the features of a swap contract with example.

Q.3. (a) Difference between Future and Option

OR

(b) “Derivatives are considered as risk management tools used by organizations/ investors/ individuals”. Comment upon the statement.

Q.4. (a) Assume that a trader sold short ABC stock in March by borrowing 300 shares and selling them for \$ 60/share. Calculate the net profit from the short sale assuming the trader bought back the shares in June for 50/ share to replace the borrowed shares and close out his short position.

OR

(b) Explain the term “mechanism of stop loss”. How it can be used to hedge the risk resulting from writing an option. Support your answer with a suitable example.

Q.5. (a) A long forward contract is equivalent to a long position in call option and a short position in put option". Explain this statement.

OR

(b) Does a perfect hedge always succeed in locking in the current spot price of an asset for a futures transaction? Explain your answer.

SECTION-C

(Case study)

Q.6. Attempt all parts of the following:

(2×15=30)

Practical Questions

6 (a) Current market price of the shares of Go- Cool Ltd. is Rs 200 and a call option with exercise price of 220 with three months to expiration is available. It is expected that spot price of these shares at the end of three months from now might increase by 40% of the current spot price or it might decline by 20% of the current spot price. Calculate delta- hedge ratio.

6 (b) Mr. X believe that the price of a particular underlying, currently selling at \$ 99, is going to increase substantially in the next six months. He decides to purchase a call option expiring in six months on this underlying. The call option has an exercise price of \$ 110.

(a) Determine the payoff if the price of the underlying six months later is \$99.

- A \$ 7
- B \$ 0
- C - \$ 8.5

(b) Determine the payoff if the price of the underlying six months from now is \$115

- A \$ 8
- B \$ 0
- C \$ 3